The Science of Sound

Science • Grades 5–8 • Red Level

🔴 **Red Level: Go Deeper**

Name: ______________________________________ Date: _______________ Period: _______

Challenge Context: The Harlem Renaissance was one of the most significant cultural events in American history — and one of the least understood. It was not simply a celebration of Black talent; it was a deliberate, sophisticated, and sometimes contested attempt to use cultural achievement as a tool for social and political change. The figures who made it were artists AND activists, creators AND strategists.

ACTIVITY 1: THE PHYSICS OF IMPROVISATION

Jazz as Applied Wave Science

Jazz improvisation — playing spontaneously within a harmonic structure — requires instant, intuitive understanding of frequency relationships. Musicians work within keys (sets of related frequencies) and chord progressions (sequences of simultaneous frequencies that create consonance or dissonance).

Research and explain: What is the relationship between musical consonance and the frequency ratios of notes? Why do some combinations of frequencies sound "good" together while others create tension?

🔴 **Red Level: Go Deeper**

ACTIVITY 2: DESIGN AN EXPERIMENT

Testing Frequency Perception

Design a scientific experiment to test one of the following questions. Include hypothesis, method, variables, and how you would measure results:

- Does tempo affect emotional response to music?
- Can people identify instruments by frequency range alone?
- How does amplitude affect perceived distance of a sound?

Question I chose:

Hypothesis:

Method (brief):

Variables (independent / dependent / controlled):

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ACTIVITY 3: THE NEUROSCIENCE OF MUSIC

Why Does Music Move Us?

Music activates multiple brain regions simultaneously — auditory cortex, limbic system (emotion), motor cortex (movement), and prefrontal cortex (expectation and prediction). Jazz improvisation, in particular, deactivates the dorsolateral prefrontal cortex (associated with self-monitoring) and activates the medial prefrontal cortex (associated with self-expression).

What does this neuroscience suggest about why jazz improvisation feels different from reading from sheet music? What might this tell us about creativity more broadly?

Critical Question: The Harlem Renaissance musicians were working intuitively with principles of physics and neuroscience they had no formal training in. What does this suggest about the relationship between scientific knowledge and artistic mastery?

🏛 Mini Museum Connection

Option 1 — Visit the Exhibition: If the Harlem Renaissance exhibition is available at your school, visit the Mini Museum to experience the phonograph, Jazz Lab, Great Migration Map, and more. Ask your librarian to arrange a class visit.

Option 2 — Virtual Visit: Explore the full exhibition at **minimuseumproject.com/exhibitions/harlem-renaissance**

Research Extension: Investigate "absolute pitch" (perfect pitch) — the ability to identify a musical note without a reference tone. What percentage of people have it? Is it genetic? Is it more common in people who learned music before age 7? Present your findings.

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